MCQS

CS312 midterm preparation file

ZB

بسم الله الرحمن الرحيم

In the name of God , Most Gracious, Most merciful

CS312

MCQS

1. What is the time complexity of accessing an element in an array?

a) O(1)

b) O(log n)

c) O(n)

d) O(n log n)

Answer: a) O(1)

2. Which of the following is the correct definition of a linked list?

a) A data structure with elements in contiguous memory locations.

b) A sequence of elements in which each element points to the next one.

c) A collection of data items of the same type.

d) A data structure that allows random access.

Answer: b) A sequence of elements in which each element points to the next one.

3. Which algorithm is used to find the shortest path in a weighted graph?

a) Dijkstra's Algorithm

b) Binary Search

c) Merge Sort

d) Breadth-First Search

Answer: a) Dijkstra's Algorithm

4. What is the space complexity of merge sort?

a) O(1)

b) O(n)

c) O(n log n)

d) O(log n)

Answer: b) O(n)

MCQS

Midterm preparation file

ZB

5. Which of the following sorting algorithms has the best average case time complexity?

a) Bubble Sort

b) Merge Sort

c) Quick Sort

d) Selection Sort

Answer: c) Quick Sort

6. In a binary search tree, which of the following is true for the left child of a node?

a) It contains a value greater than the parent node.

b) It contains a value less than or equal to the parent node.

c) It contains a value equal to the parent node.

d) It contains a value greater than or equal to the parent node.

Answer: b) It contains a value less than or equal to the parent node.

7. What does the term 'polynomial time' mean?

a) Time complexity grows faster than any polynomial function.

b) Time complexity grows at a rate that is a polynomial function of input size.

c) Time complexity is linear.

d) Time complexity grows exponentially.

Answer: b) Time complexity grows at a rate that is a polynomial function of input size.

8. Which of the following is not a characteristic of a stack?

a) LIFO (Last In First Out)

b) Supports push and pop operations

c) Random access of elements

d) Supports peek operation

Answer: c) Random access of elements

9. What is the time complexity of binary search?

a) O(n)

b) O(log n)

c) O(n log n)

d) O(1)

Answer: b) O(log n)

10. Which type of tree has all of its levels completely filled except possibly the last, which is filled from left to right?

a) Binary Search Tree

b) AVL Tree

c) Complete Binary Tree

d) Red-Black Tree

Answer: c) Complete Binary Tree

11. What is the worst-case time complexity of quick sort?

a) O(n)

b) O(n log n)

c) O(log n)

d) O(n^2)

Answer: d) O(n^2)

12. Which data structure is used for implementing recursion?

a) Queue

b) Stack

c) Array

d) Linked List

Answer: b) Stack

13. Which of the following is a non-linear data structure?

a) Array

b) Linked List

c) Stack

d) Tree

Answer: d) Tree

14. Which of the following is not a valid graph traversal technique?

a) Depth-First Search

b) Breadth-First Search

c) Quick Sort

d) Dijkstra's Algorithm

Answer: c) Quick Sort

15. What is the time complexity of insertion sort in the worst case?

a) O(n)

b) O(log n)

c) O(n log n)

d) O(n^2)

Answer: d) O(n^2)

16. What is the space complexity of quicksort in the best case?

a) O(1)

b) O(n)

c) O(n log n)

d) O(log n)

Answer: d) O(log n)

17. In a binary heap, the parent node is greater than or equal to its children. This property is called:

a) Min-Heap Property

b) Max-Heap Property

c) Heap Property

d) Priority Queue Property

Answer: b) Max-Heap Property

18. Which sorting algorithm is generally the most efficient for large datasets?

a) Bubble Sort

b) Merge Sort

c) Insertion Sort

d) Quick Sort

Answer: d) Quick Sort

19. Which data structure does Dijkstra's algorithm use for finding the shortest path?

a) Queue

b) Stack

c) Priority Queue

d) Binary Tree

Answer: c) Priority Queue

20. What is the space complexity of the depth-first search (DFS) algorithm?

a) O(1)

b) O(n)

c) O(log n)

d) O(n^2)

Answer: b) O(n)

MCQS Midterm preparation file

ZB

21. Which of the following algorithms is a divide-and-conquer algorithm?

a) Merge Sort

b) Selection Sort

c) Insertion Sort

d) Quick Sort

Answer: a) Merge Sort

22. Which is not a feature of a binary search tree (BST)?

a) Left child < Parent

b) Right child > Parent

c) Parent is greater than all of its children

d) Inorder traversal gives sorted output

Answer: c) Parent is greater than all of its children

23. What is the time complexity of accessing an element in a hash table on average?

a) O(n)

b) O(log n)

c) O(1)

d) O(n log n)

Answer: c) O(1)

24. Which of the following is an advantage of a doubly linked list over a singly linked list?

a) Faster insertions and deletions

b) No extra memory needed for the previous pointer

c) Can be traversed in both directions

d) More memory efficient

Answer: c) Can be traversed in both directions

Midterm preparation file

25. Which of the following sorting algorithms has a best-case time complexity of O(n)?

a) Merge Sort

b) Bubble Sort

c) Insertion Sort

d) Quick Sort

Answer: c) Insertion Sort

26. Which of the following is true about a priority queue?

a) It supports random access

b) It always returns the smallest element

c) It returns elements based on their priority

d) It is implemented using an array

Answer: c) It returns elements based on their priority

27. What is the time complexity of the delete operation in a binary heap?

a) O(1)

b) O(log n)

c) O(n)

d) O(n log n)

Answer: b) O(log n)

28. Which type of graph is used in breadth-first search (BFS)?

a) Directed Graph

b) Undirected Graph

c) Weighted Graph

d) Any type of graph

Answer: d) Any type of graph

Midterm preparation file

29. What is the time complexity of the merge operation in merge sort?

a) O(n)

b) O(log n)

c) O(n log n)

d) O(n^2)

Answer: a) O(n)

30. Which of the following is an example of a greedy algorithm?

a) Dijkstra's algorithm

b) Merge Sort

c) Quick Sort

d) Binary Search

Answer: a) Dijkstra's algorithm

31. Which data structure is used in implementing the depth-first search (DFS) algorithm?

a) Queue

b) Stack

c) Array

d) Linked List

Answer: b) Stack

32. Which of the following is true about hash tables?

a) They always have a constant time complexity for operations

b) They use linked lists to resolve collisions

c) They are inefficient for lookups

d) They are ideal for searching and inserting data

Answer: d) They are ideal for searching and inserting data

Midterm preparation file

33. Which of the following algorithms uses a divide-and-conquer approach to solve the problem?

a) Merge Sort

b) Selection Sort

c) Heap Sort

d) Insertion Sort

Answer: a) Merge Sort

34. In a hash table, what happens if two keys hash to the same index?

a) The second key is discarded

b) Collision occurs, and it must be handled

c) The table automatically resizes

d) A new table is created

Answer: b) Collision occurs, and it must be handled

35. Which of the following operations is not supported by a queue?

a) Enqueue

b) Dequeue

c) Peek

d) Random Access

Answer: d) Random Access

36. Which of the following is the time complexity for the best case of quick sort?

a) O(n^2)

b) O(n log n)

c) O(n)

d) O(log n)

Answer: b) O(n log n)

Midterm preparation file

37. Which of the following is the correct approach for breadth-first search (BFS)?

a) Stack

b) Queue

c) Priority Queue

d) Array

Answer: b) Queue

38. What is the maximum number of nodes at level 'l' in a binary tree?

a) 2^l

b) 2^l - 1

c) 2^(l+1)

d) 2^(l-1)

Answer: a) 2^l

39. Which of the following is the time complexity of searching in a balanced binary search tree (BST)?

a) O(n)

b) O(log n)

c) O(n log n)

d) O(1)

Answer: b) O(log n)

40. What is the space complexity of a recursive algorithm with a stack of size N?

a) O(1)

b) O(log N)

c) O(N)

d) O(N^2)

Answer: c) O(N)

ZB

MCQS

Midterm preparation file

41. Which of the following is a feature of a heap data structure?

a) It is always a balanced binary tree.

b) The children are always smaller than the parent node.

c) It supports fast insert and delete-min operations.

d) It uses an array for storage.

Answer: c) It supports fast insert and delete-min operations.

42. What is the time complexity of accessing an element in a linked list?

a) O(1)

b) O(log n)

c) O(n)

d) O(n log n)

Answer: c) O(n)

43. Which data structure does the depth-first search (DFS) use to store nodes to be explored?

a) Queue

b) Stack

c) Array

d) Hash Table

Answer: b) Stack

44. In an AVL tree, what condition must be true for the tree to be balanced?

a) The difference between the heights of the left and right subtrees must be at most 1.

b) The left and right subtrees must be equal in height.

c) All nodes must be leaf nodes.

d) The root node must have no children.

Answer: a) The difference between the heights of the left and right subtrees must be at most 1.

45. Which of the following statements is true about a binary search tree (BST)?

a) It is a type of balanced tree.

b) It allows for random access to nodes.

c) Inorder traversal produces sorted data.

d) It supports only the insertion operation.

Answer: c) Inorder traversal produces sorted data.

46. What is the time complexity of bubble sort in the worst case?

a) O(n)

b) O(n log n)

c) O(n^2)

d) O(log n)

Answer: c) O(n^2)

ZB +92 300 5427579

WhatsApp:

https://whatsapp.com/channel/0029VaODheTDp2Q8YVcoON2p

Second WhatsApp channel

channel on WhatsApp:

https://whatsapp.com/channel/0029VafvOcy90x2tPbUaX02I

YouTube:

https://youtu.be/-AMj4iNz0UE?si=hg29pJZ1kujzg57J

Phone no : +92 300 5427579

Phone no: +92 330 3869586

Email: asktozb@gmail.com

Passing criteria of this course

Letter Grade	Grade Points	Equivalent Percentage
A+	4.00	90-100
A	4.00	85-89
A-	3.66 - 3.99	80-84
B+	3.33 - 3.65	75-79
B	3.00 - 3.32	71-74
B-	2.66 - 2.99	68-70
C	2.00 - 2.65	61-67
D	1.00 - 1.99	50-60
F	0.00	00-49

Passing Criteria for a Course:

The passing criteria is defined in such a way that it ensures the student shall be consistent in his studies throughout the semester.
Therefore, for passing a course/subject, student shall fulfill the following:
a) Secure minimum 20% score in Formative Assessments/Mid terms
b) Secure minimum 20% score in Final Term Examinations.
c) Secure at least 40% marks in aggregate while fulfilling the above requirements

Course Selection and Credit Hours

Q: How do I select courses when the course selection is open?
A: Follow these steps:
1. Check the credit hours allowed by your university (e.g., 21, 18, or 15 credit hours).
2. Each subject typically has 3 credit hours.
3. Divide the total allowed credit hours by 3 to determine how many subjects you can select.
Examples:
21 credit hours = 7 su18 credit hours = 6 subjects
15 credit hours = 5 subjects
Remember, the university may allow different credit hours for each student, so check your specific allowance.
By following these steps, you can make informed decisions during course selection and manage your credit hours effectively.

ZB

Advice

Hmaesha relax ho ke parha kryn panic mat hua kryn apky parents ky bohat khawab hoty hn wo pura krny ki puri koshish kea kryn kbi b ksi pe depend na kea kryn apko bs ak insan success kr saqta ha or wo insan ap khud hn apky elawa koi nahe ap bs koshish kryn Allah pak pe strong yakeen rakha kryn or mehnat krke sb Allah ke hawaly kr dea kryn everything is possible be brave be strong stay blessed
Hum insan hamesha moat se darty or bhagty hn or moat se bachny ki koshish krty hn jabke moat ny ana hi ana ha hum

moat se nahe bach saqty humy jahanam se bachny ki koshish krni chaheay hum jahanam se bach saqty hn
Insan ko 3 chezo se dar lagta ha Moat Risk/Dolat Ezat Shohrat Fame

Moat

Humy pta hona chaheay Moat tab ani ha jab Allah chahy us se pehly puri dunya bi ak tararf ho ke apko marna chahy to apka kuch nahe beggar saqti

Risk/Dolat

Risk Dolat sb Allah pak ke hath m ha puri dunya ak taraf b ho jay na apsy apka risk cheen saqti ha na de saqti ha Ye sirf ALLAh pak ky hath m ha

Izat Zilat shohrat Fame

Izat Zilat Sirf Allah de saqty hn puri dunya mel ke b apko zra brabr b demage nahe kr saqti Agr ALLAH

Apko izat dena chahay to puri dunya ak tararf ho ke b apko 1% b nuksan nahe pohancha saqti
Phr Dar ks bat ka ??????????????
Be brave be strong Just put your trust To ALLAH
Hamesha Confident or himat se raho kbi ksi k samne mat jhuko puri dunya mel k b apka kuch b nahe beggar saqti

ZB
MY REQUEST FROM ALL OF YOU

ZB Request from All of you My family
Mjy ap sbki help or support ki zarort ha or wo ye k hum sb Mel kr Allah ka Quran ki Urdu translation logo tak pohanchy or Quran ko samjna asaaan kryn dosro k leay sb tak Allah ka Quran pohanchy it's my compaign hum log 70+ age k ho jaty hn phr b hmy namaz tak ki translation nahe pta hoti k hum Allah pak se

Kya Dua kr rhy hn so hum youngster's ko Mel kr puri takat or energy ke sath Allah ka Quran spread krna ha
Hum puri Koshish kr rhy hn k hum sb Mel k Quran ki translation or most important topics ko maximum share kry with translation hmara maksad Quran ko spread krna ha be a part of us
m apni pocket se ye sb kr rha ho Allah ka Quran spread krny k leay apne or mene sbny Marna ha ak din to q na Allah k leay Kuch kryn apni energies apni power ko bajay negative use krny k Allah k leay invest kryn khud ko be a part of us
Agr m 23 ki age m apny sare sources use kr rha ho pocket se heavy amounts give away kr rha ho Quran spread krny k leay to ap just share to kr saqty hn itna e krdyn

Contact ZB +92 300 5427579

ZB
Agr ap mujsy contact na kr sky too b ap jo b hn jaha bi hn waha Allah ka quran pheelay zada se zada with urdu translation jitni himat ha utna share kryn

MY Family

ust Relax and focus! Exams are not difficult. Put in your 100% effort and trust in Allah.

AL-Quran

"And indeed, with hardship comes ease." (Quran 94:5)
Remember, I'm here to support you! Stay brave and strong!"

May Allah bless you